Emotion Recognition from Physiological Signals using Time-Frequency Analysis Methods

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Abstract**—**Emotions play an important role in the psychology and physiology of the human. It originates from the day to day experiences, people they interact with and the environment they deal with. In the current day of technological development, emotion recognition has got great attention, since it can contribute to the Human-machine interaction and for complete automation. Emotion recognition from physiological signals can help people with impairments in social interaction, communication and developmental disorders. This work tries to detect the human emotions from physiological signal, Electrocardiogram (ECG) using time-frequency analysis methods. It also provides a comparison of the performance of different time frequency methods of feature extraction. Discrete wavelet transform and Hilbert Haung transform are the time frequency methods used. For comparison two classification techniques Support Vector Machine and K-Nearest Neighbhour are also used. The emotions considered are happy and sad. Database was obtained from DECAF, a database of all physiological signals of different emotions.

Keywords—Ecg ,Dwt, Hht,St.

I. INTRODUCTION

Emotion is the complex state of behaviour that influence thought and behaviour. Emotion consists of the subjective component which describes how we experience emotions, a physiological part that corresponds to how our body reacts to the emotions and an expressive component which reflects in the response of the human to each emotion. The term emotion refers to relations among external incentives, thoughts, and changes in internal feelings. Emotion recognition has evolved into a vast area of research in cognitive science engineering and psychology. Emotions of a person can be identified using various techniques. By analyzing facial expressions, gestures, eye movement and by analyzing and processing the speech. The drawback of these methods is that one can disguise their emotions and hide the real emotions. Physiological detection of emotions can have better performance compared to other techniques since physiological signals are not under the voluntary control of the human. Emotion can be modeled as two dimensional emotion model consisting of valence and arousal, where valence denotes pleasantness or polarity of emotion stimuli whereas arousal represents the strength of emotion. Physiological signals like EEG, PPG, GSR EMG... can also be used to recognize emotions. Since brain and emotions are not mapped for autistic people EEG cannot be used for emotion recognition for such people. Here in this work ECG is employed for detecting the emotions. Time frequency methods are used to decompose the signal and features are extracted from the decomposed signal. High frequency and low frequency power are the features extracted. These extracted features are used as training data for the classifier. SVM and KNN are the classifiers used in this work. This paper consists of 5 sections. Section II describes about the previous works done in the field. It gives a detailed review of works carried out in this field. Section III describes about the proposed system and section IV gives the results and discussion and section V gives the conclusion of the work.

II. RELATED WORK

Ofer Golan et al. [6] had done experiments to study emotional features of children coming under Autism Spectrum Disorder. He studied emotion identification and regulation features by conducting experiments using 20 autistic children of the age from 4-7. He used 'The Transporter' an animation movie to induce emotions in them. He identified from his study that due to less attention of such children towards faces and eyes they have impairment in recognizing emotions and in face processing skills.

Many people took the idea of multimodality i.e. using more than one physiological signal for analysis. Better result was obtained for such research works. The main steps in emotion recognition from physiological signal are pre-processing of the signal to remove noise, feature extraction, feature selection (to eliminate those features which does not contribute much for emotional information) and classification. Several papers in this field make use of different methods in each of the steps. The work done by Sasikumar K et al. [8] was meant for identifying physiological signal variations between control group and autistic group. The experiment was conducted using 40 normal and 40 autistic children. They used clue cards containing alphabets, animals, colours and fruits to elicit emotions and acquired PPG and GSR signals from people under test. During the mental ability task it was seen that autistic children are not getting concentration level as that of control group. They become more stressed and have an increased ANS activity. As a result they showed an increased

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heart rate. The results of the paper shows that the PPG and GSR signal recorded are having high values for autistic children when compared to normal children.

A physiological reaction is a change in activity in the autonomic nervous system (ANS) that accompanies emotions. Murugappan et al.[2] tried to analyze and classify 5 different emotions happiness, disgust, fear, sadness and neutral from HRV signals derived from ECG. DWT is used to extract statistical features from HRV signal. Four types of wavelet functions daubchies6(db6), daubchies7(db7), Symmlet8 (sym8), and Coiflet5 (coif5) are used. LDA and KNN are the classification technologies used. Fourteen level decomposition is done to obtain low frequency(0.03-0.12HZ) and high frequency(0.12-0.488 Hz) which contains emotional information. Of these KNN provided maximum classification accuracy compared to LDA.

Bio signals manifest directly underlying activity of the autonomous nervous system. The paper proposed by Jerritta et al. [3] use 6 emotional states happiness, sadness, fear, surprise, disgust and neutral for classification. Emotion is induced in the subjects using audio and visual clips. Hilbert Huang Transform (HHT) based analysis was done to obtain the emotional features in low and high-frequency ranges. Emotional data are more found in low-frequency intrinsic mode function than in other frequencies.
Krupa et al.[4] developed a wristband for acquiring physiological signals and the physiological reaction is found by variation in parameters like GSR and HRV. This paper focuses on the emotion recognition of autistic children so that it may help caretakers to provide them proper care during the early developmental stages.

Emotions surprise, boredom and pain are recognized and classified using DFA (the statistical method), and five machine learning algorithms LDA, CART, SOM, SVM and Naive Bayes by Eun-Hye Jang et al.[5] ECG, EDA, SKT and PPG are the physiological signals used for the analysis of these emotions.

The paper proposed by Jin Hun Sohn[1] is a comparative study of the efficiency of different classification methods. LDA, CART, SVM and SOM are the classification methods used. The efficiency of these classifications methods is found by classifying 3 different emotions boredom pain and surprise. It has considered four physiological signals EDA, ECG, PPG and SKT. Physiological signals are used for emotion recognition in individuals because acquiring these signals can be done non invasively using some sensors. Signals are acquired for a time duration of 1-1.5 minutes during emotional state. 23 features are extracted from all the physiological signals and are classified. Of that maximum accuracy is obtained for SVM classifier.

DECAF[10] is a database containing the physiological response to different emotions elicited in 30 subjects using 36 movie clips and 40 one minute music video segment. They had collected different signals such as MEG, horizontal electrooculogram hEOG, ECG, trapezium electromyogram (tEMG) and near-infrared facial videos which were recorded synchronously. Hao Min[9] proposed a method of classifying two emotions joy and sadness from ECG signal. 391 subjects participated in the experiment. Emotion was stimulated using film clips. BIOPAC MP150 was used for acquisition of ECG signal. emotion was detected and classified by extracting HRV parameters and several statistical features such as standard deviation of S wave amplitude, maximum of R wave amplitude, mean of S wave amplitude etc. Tabu search algorithm was used for feature selection to eliminate those features which are less relevant for emotion recognition from ECG signal.

Physiological responding of the human being as a result of his psychological reaction, human emotional states can be easily derived from his physiological signals. A study of physiological signal relation to emotion is done by [7] K Kiruba et al. They observed variation ECG signal, skin conductance, respiration rate and skin temperature with respect to emotional changes and changes in mental states. The study used AIEFS for feature selection and HEC for estimation evaluation. Support vector regression with genetic algorithm, multinomial naive Bayes, ensemble online sequential extreme learning machine are used in a combined form for classification to get increased performance.

III. METHODOLOGY

The work gives a performance comparison between two time frequency analysis methods and two classifiers. Time frequency analysis methods are discrete wavelet transform and Hilbert haung transform. KNN is the classifier used. ECG data for various emotions are collected from DECAF [10]. In this work ECG signal corresponding to two different emotions happy and sad are considered.

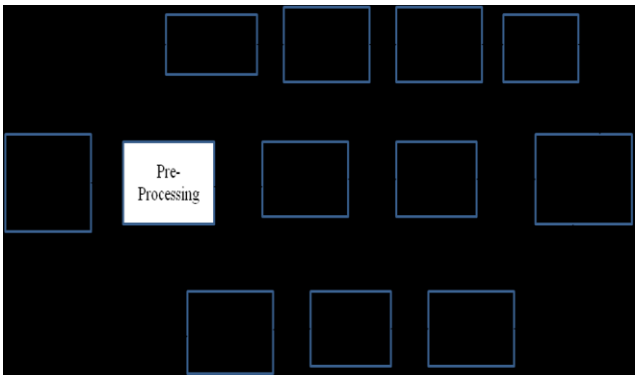

Fig. 1 Block Diagram

A. ECG Data

ECG data was collected from DECAF[10]. It is a multimodal database containing physiological response to various emotions. The emotions are elicited using 36 one minute movie segments and 40 one minute music segments. The experiment was carried out for 30 healthy subjects. In this work only ECG signal response to the one minute movie segment for emotions happy and sad was analysed.

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B. Pre-Processing

The obtained ECG signal is corrupted with large amount of noise. So the first step in processing ECG signal is to remove the artefacts. Different types of noises corrupting ECG signal are power line interference, baseline wander and high frequency noise. In India since the power line frequency is at 50 Hz, a notch filter at 50 Hz can remove power line interference.. For removing high frequency noise a low pass filter at 100Hz was used.

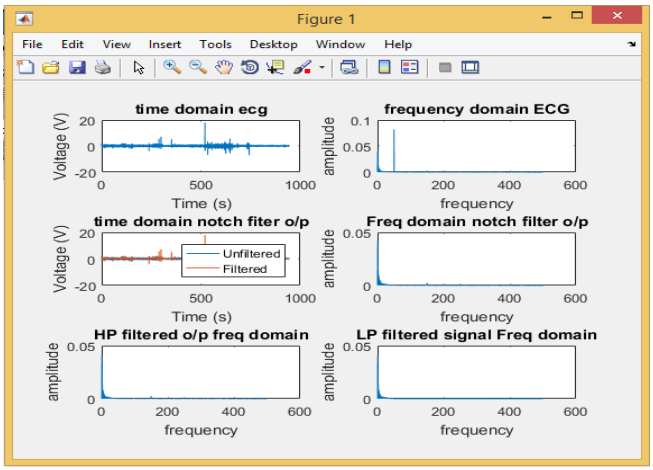


Fig. 2 Output ECG after Pre-Processing

The main reason for High frequency noise is that the instrumentation amplifiers or recording system may pick up the ambient EM signals by cables. A low pass filter with a cut off frequency of 70 Hz will remove high frequency noise components. Base line interference and low frequency noise is caused mainly due to breathing or coughing with large movement of chest in chest lead ECG and due to the movement of leg or arm in limb lead ECG. Variation in temperature and bias in instrumentation amplifier also causes base line drift. It can be removed by using a high pass filter with cut off frequency 2 Hz. The noise free ECG signal is used for further steps.

This work attempts to detect the emotions of a person by analysing ECG signal of the subject. Here time-frequency analysis of the ECG signal is done. Three different time frequency methods are employed and comparison between the three methods are also done. Discrete Wavelet Transform, Hilbert Haung Transform and Stockwell transform are the Time Frequency analysis methods used.

C. *Signal decomposition & feature extraction*

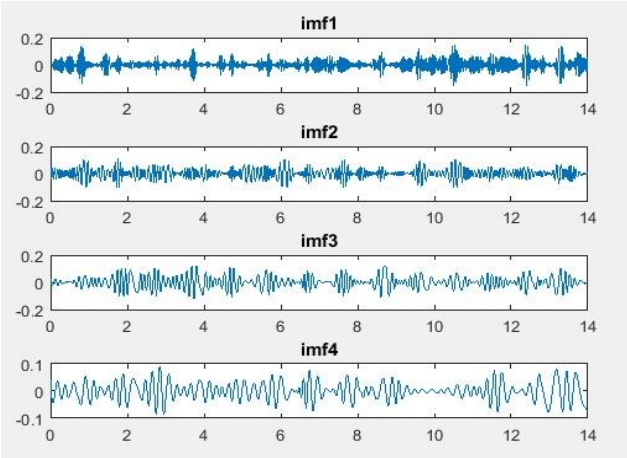
DWT:DWT is a time frequency analysis method which has good time and frequency resolution. Size of the window adapts to the frequency of the signal such that as the frequency of the signal increases the size of the window decreases and as the frequency of the signal decreases the size of the window increases. Since the emotional data is present in the low frequency (.03-0.12Hz (LF), 0.12-0.488 Hz (HF)and high frequency of ECG signal, to extract these components a 14 level decomposition is done using wavelet transform. 11th, 12th, 13th and 14th level coefficients are used for feature extraction.

TABLE. 1 Frequency Bands and Decomposition Levels

Frequency range(Hz)	Decomposition level	Frequency Bands
0.03-0.06	D14	Low frequency band
0.06-0.12	D13	
0.12-0.24	D12	High frequency band
0.24-0.488	D11	
0.48-0.976	D10	Unused frequency bands
0.976-1.953	D9	
1.9532-3.90625	D8	
3.90625-7.8125	D7	
7.8125-15.625	D6	
15.625-31.25	D5	
31.25-62.5	D4	
62.5-125	D3	
125-250	D2	
250-500	D1	

Low frequency power, high frequency power, total power and ratio of HF power to LF power are the features extracted. Mean, standard deviation, minimum value, maximum value of last 4 level detail coefficients are also taken as features. Db6, Db7, sym8, coif5 are wavelets taken in this work. Performance comparison of these wavelets are also done.

HHT: Hilbert haung transform is a combination of empirical mode decomposition and Hilbert transform. Every signal consists of a number of co-existing frequencies one overlapped on the other. EMD decomposes a signal in to its component frequencies and produces intrinsic mode functions. Hilbert transform of these intrinsic mode functions are found out. HF Power, LF Power and total power are the features extracted.



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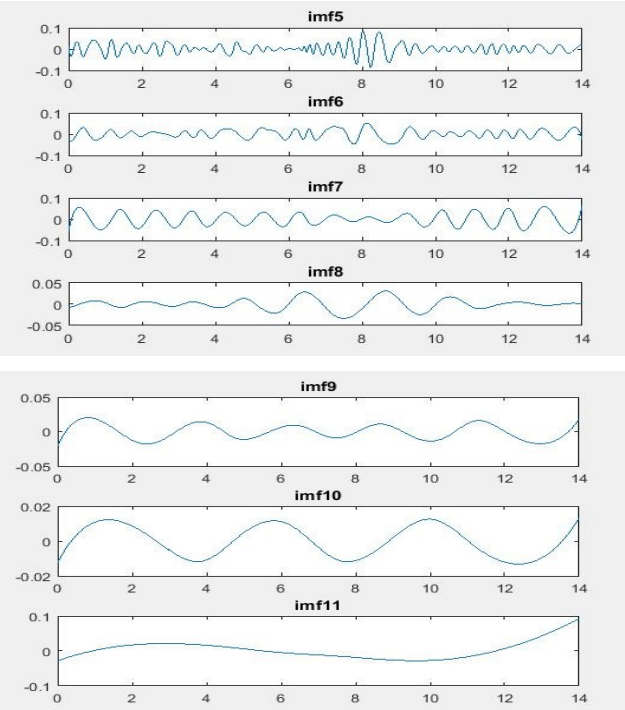

Fig. 3 Intrinsic Mode Functions

ST: Stockwell Transform is a time-frequency analysis method combining the advantages of both short time fourier transform and wavelet transform. Since it uses frequency dependent Gaussian window, its frequency resolution increases. In ST oscillatory exponential kernel is stationary and Gaussian window is translated, retaining the absolutely referenced phase. ST S(□,f) is defined as the product of fourier transform of x(t) and Gaussian window.

$$S(\tau,f)=\frac{|f|}{2\pi}\int_{-\infty}^{\infty} x(t)\, e^{-\frac{(t-\tau)f^2}{2}} e^{-i2\pi f t} dt$$

Width of gaussian window is given by

$$\sigma(f) = T = \frac{1}{|f|}$$

D. Classifiction

ECG signal of 10 subjects with the emotions happy and sad from the database are selected for feature extraction. KNN classifier is used for emotion classification. The extracted features are given as training data for the classifier.

IV. RESULTS AND DISCUSSION

TABLE. 2 Results

Method	Experiment type	Accuracy(%)		Precision(%)		Specificity(%)		Sensitivity(%)	
		Happy	Sad	Happy	Sad	Happy	Sad	Happy	Sad
	Subjective	55	55	55.56	54.55	60	50	50	60
coif5 (DWT)	Non subjective	60	60	60	60	60	60	60	60
sym8 (DWT)	Subjective	57.50	57.50	57.14	57.89	55	60	60	55
	Non subjective	55	55	54.17	56.25	45	65	65	45
db6 (DWT)	Subjective	67.50	67.50	68.42	66.67	70	65	65	70
	Non subjective	50	50	50	50	60	40	40	60
db7 (DWT)	Subjective	75	75	77.8	72.73	80	70	70	80
	Non subjective	57.50	57.5	57.89	57.14	60	55	55	60
HHT	Subjective	67.5	67.5	81.82	62.07	90	45	90	45
	Non subjective	65	65	68.75	62.5	75	55	55	75
ST	Subjective	60	60	66.67	57.14	80	40	40	80
	Non subjective	65	65	68.75	62.50	75	55	55	75

V. CONCLUSION

Subjective analysis provided better performance when compared to non subjective analysis. Db7 has got more accuracy in recognizing human emotions from ECG signal in subjective analysis while comparing with all wavelets in DWT as well as with the technique using HHT. In non subjective analysis HHT and ST performed well.

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